

**DECISION MAKING SYSTEM
ANALYTICAL HIERARCHY PROCESS (AHP)**



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A. Background

A manufacturing company is looking for the best supplier to procure its main raw materials. This decision is very important because it will affect production costs, product quality, and operational efficiency. For this reason, the company uses the Analytical Hierarchy Process (AHP) method as a Decision Support System (SPK) in determining the best supplier.

B. Goals

Determine the best supplier that provides maximum benefits at minimal cost.

C. Breakdown Process

In this assignment, we analyze and break down the criteria into 2, namely benefits and costs. The following are the criteria based on these categories:

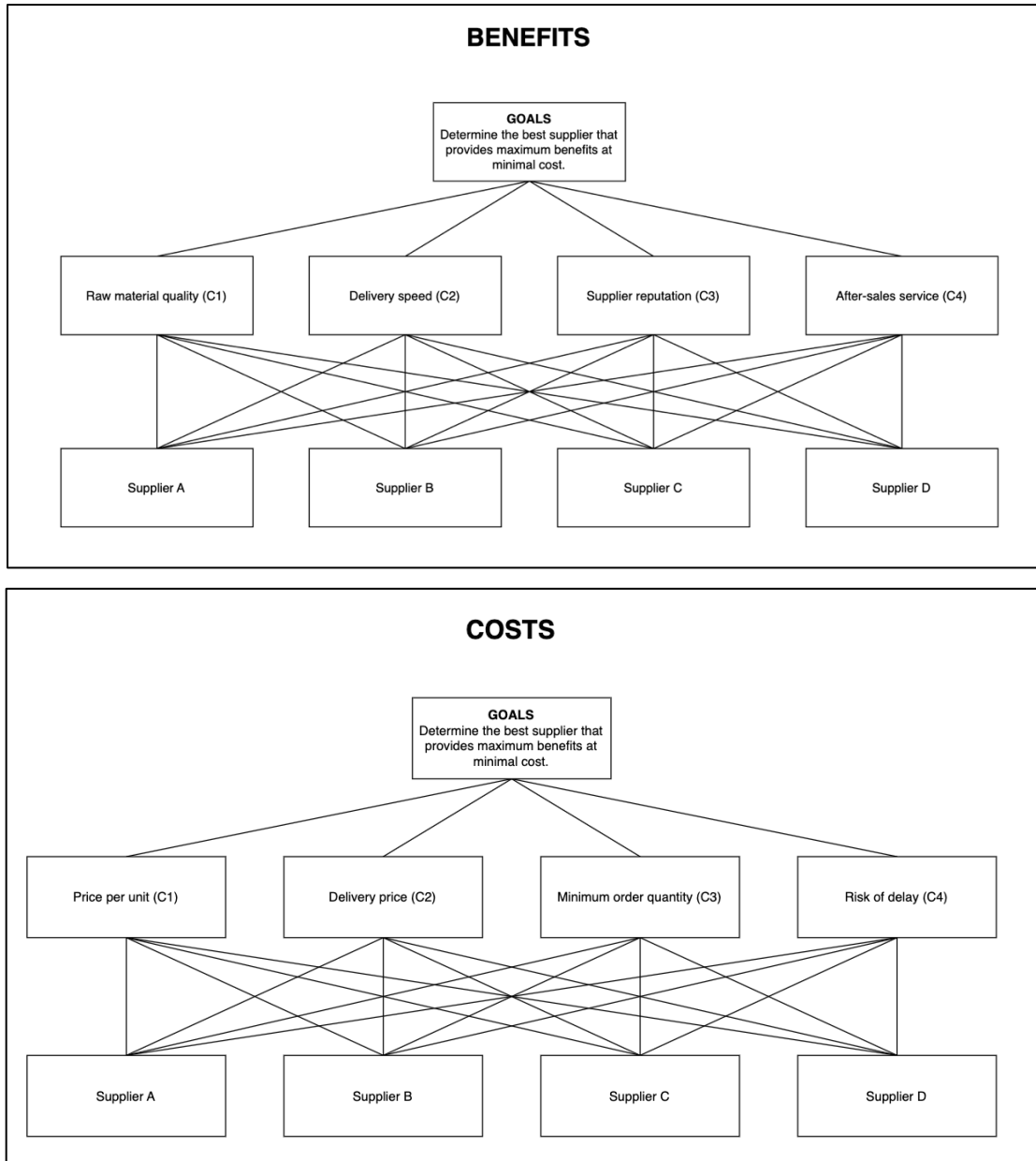
- **Benefits** (The higher the score, the better)
 - **Raw Material Quality:** Standard raw materials that meet company specifications.
 - **Delivery Speed:** The time it takes for suppliers to deliver orders.
 - **Supplier Reputation:** The supplier's credibility and track record in the industry.
 - **After-Sales Service:** Technical support, warranty, or replacement of problematic products.
- **Costs** (The lower the value, the better)
 - **Price per Unit:** The cost per unit of raw materials offered by the supplier.
 - **Shipping Cost:** Logistics and transportation costs that must be incurred by the company.
 - **Minimum Order Quantity:** The minimum amount that must be purchased in one transaction.
 - **Risk of Delay:** Possible delays that could affect production.

For alternatives, we have 4 alternatives:

- Supplier A
- Supplier B
- Supplier C
- Supplier D

D. Hierarchy of Problems

After we find the criterias and alternatives, we then draw a hierarchy diagram as follows:



E. Benefits Calculation

- Intensity of Interest

Intensity of Interest				
Criteria	C1	C2	C3	C4
C1	1/1	3/1	3/1	5/1
C2	1/3	1/1	1/1	3/1
C3	1/3	1/1	1/1	3/1
C4	1/5	1/3	1/3	1/1

- Normalization

Normalization				
Criteria	C1	C2	C3	C4
C1	1,00	3,00	3,00	5,00
C2	0,33	1,00	1,00	3,00
C3	0,33	1,00	1,00	3,00
C4	0,20	0,33	0,33	1,00
Total	1,87	5,33	5,33	12,00

Criteria	C1	C2	C3	C4	Total	PW
C1	0,54	0,56	0,56	0,42	2,08	0,52
C2	0,18	0,19	0,19	0,25	0,80	0,20
C3	0,18	0,19	0,19	0,25	0,80	0,20
C4	0,11	0,06	0,06	0,08	0,32	0,08

- Calculating Ratio

Calculating Ratio							
Criteria	C1	C2	C3	C4	PW	Matrix Results	Divided Results
C1	1,00	3,00	3,00	5,00	0,52	2,12	4,08
C2	0,33	1,00	1,00	3,00	0,20	0,81	4,04
C3	0,33	1,00	1,00	3,00	0,20	0,81	4,04
C4	0,20	0,33	0,33	1,00	0,08	0,32	4,02

- Count Maks

Count Maks		
Total	4,04	

- Count Consistency Index

Count Consistency Index		
Total	0,01	

- Consistency Ratio

Consistency Ratio		
Total	0,02	Consistent

- Consistency Ratio for C1

Intensity of Interest - C1						
Criteria C1	Supplier A	Supplier B	Supplier C	Supplier D		
Supplier A	1/1	1/1	3/1	2/1		
Supplier B	1/1	1/1	2/1	1/1		
Supplier C	1/2	1/2	1/1	1/1		
Supplier D	1/2	1/1	1/1	1/1		
Normalization - C1						
Criteria C1	Supplier A	Supplier B	Supplier C	Supplier D		
Supplier A	1,00	1,00	3,00	2,00		
Supplier B	1,00	1,00	2,00	1,00		
Supplier C	0,50	0,50	1,00	1,00		
Supplier D	0,50	1,00	1,00	1,00		
Total	3,00	3,50	7,00	5,00		
Criteria C1	Supplier A	Supplier B	Supplier C	Supplier D	Total	PW
Supplier A	0,33	0,29	0,43	0,40	1,45	0,36
Supplier B	0,33	0,29	0,29	0,20	1,10	0,28
Supplier C	0,17	0,14	0,14	0,20	0,65	0,16
Supplier D	0,17	0,29	0,14	0,20	0,80	0,20

Calculating Ratio - C1							
Criteria C1	Supplier A	Supplier B	Supplier C	Supplier D	PW	Matrix Results	Divided Results
Supplier A	1,00	1,00	3,00	2,00	0,36	1,53	4,21
Supplier B	1,00	1,00	2,00	1,00	0,28	1,16	4,21
Supplier C	0,50	0,50	1,00	1,00	0,16	0,68	4,18
Supplier D	0,50	1,00	1,00	1,00	0,20	0,82	4,12
Count Maks - C1							
Total	4,18						
Count Consistency Index - C1							
Total	0,06						
Consistency Ratio - C1							
Total	0,07	Consistent					

- Consistency Ratio for C2

Intensity of Interest - C2						
Criteria C2	Supplier A	Supplier B	Supplier C	Supplier D		
Supplier A	1/1	2/1	3/1	2/1		
Supplier B	1/2	1/1	1/1	3/1		
Supplier C	1/3	1/1	1/1	2/1		
Supplier D	1/2	1/5	1/2	1/1		
Normalization - C2						
Criteria C2	Supplier A	Supplier B	Supplier C	Supplier D		
Supplier A	1,00	2,00	3,00	2,00		
Supplier B	0,50	1,00	1,00	3,00		
Supplier C	0,33	1,00	1,00	2,00		
Supplier D	0,50	0,20	0,50	1,00		
Total	2,33	4,20	5,50	8,00		
Criteria C2	Supplier A	Supplier B	Supplier C	Supplier D	Total	PW
Supplier A	0,43	0,48	0,55	0,25	1,70	0,43
Supplier B	0,21	0,24	0,18	0,38	1,01	0,25
Supplier C	0,14	0,24	0,18	0,25	0,81	0,20
Supplier D	0,21	0,05	0,09	0,13	0,48	0,12

Calculating Ratio - C2							
Criteria C2	Supplier A	Supplier B	Supplier C	Supplier D	PW	Matrix Results	Divided Results
Supplier A	1,00	2,00	3,00	2,00	0,43	1,78	4,18
Supplier B	0,50	1,00	1,00	3,00	0,25	1,03	4,07
Supplier C	0,33	1,00	1,00	2,00	0,20	0,84	4,11
Supplier D	0,50	0,20	0,50	1,00	0,12	0,48	4,05
Count Maks - C2							
Total	4,10						
Count Consistency Index - C2							
Total	0,03						
Consistency Ratio - C2							
Total	0,04	Consistent					

- Consistency Ratio for C3

Intensity of Interest - C3						
Criteria C3	Supplier A	Supplier B	Supplier C	Supplier D		
Supplier A	1/1	3/1	2/1	3/1		
Supplier B	1/3	1/1	3/1	1/1		
Supplier C	1/2	1/3	1/1	2/1		
Supplier D	1/3	1/1	1/2	1/1		
Normalization - C3						
Criteria C3	Supplier A	Supplier B	Supplier C	Supplier D		
Supplier A	1,00	3,00	2,00	3,00		
Supplier B	0,33	1,00	3,00	1,00		
Supplier C	0,50	0,33	1,00	2,00		
Supplier D	0,33	1,00	0,50	1,00		
Total	2,17	5,33	6,50	7,00		
Criteria C3	Supplier A	Supplier B	Supplier C	Supplier D	Total	PW
Supplier A	1,00	3,00	2,00	3,00	9,00	2,25
Supplier B	0,33	1,00	3,00	1,00	5,33	1,33
Supplier C	0,50	0,33	1,00	2,00	3,83	0,96
Supplier D	0,33	1,00	0,50	1,00	2,83	0,71

Calculating Ratio - C3							
Criteria C3	Supplier A	Supplier B	Supplier C	Supplier D	PW	Matrix Results	Divided Results
Supplier A	1,00	2,00	3,00	2,00	2,25	9,21	4,09
Supplier B	0,50	1,00	1,00	3,00	1,33	5,54	4,16
Supplier C	0,33	1,00	1,00	2,00	0,96	4,46	4,65
Supplier D	0,50	0,20	0,50	1,00	0,71	2,58	3,64
Count Maks - C3							
Total	4,14						
Count Consistency Index - C3							
Total	0,05						
Consistency Ratio - C3							
Total	0,05	Consistent					

- Consistency Ratio for C4

Intensity of Interest - C4						
Criteria C4	Supplier A	Supplier B	Supplier C	Supplier D		
Supplier A	1/1	2/1	2/1	3/1		
Supplier B	1/3	1/1	3/1	1/1		
Supplier C	1/2	1/3	1/1	2/1		
Supplier D	1/3	1/1	1/2	1/1		
Normalization - C4						
Criteria C4	Supplier A	Supplier B	Supplier C	Supplier D		
Supplier A	1,00	2,00	2,00	3,00		
Supplier B	0,33	1,00	3,00	1,00		
Supplier C	0,50	0,33	1,00	2,00		
Supplier D	0,33	1,00	0,50	1,00		
Total	2,17	4,33	6,50	7,00		
Criteria C4	Supplier A	Supplier B	Supplier C	Supplier D	Total	PW
Supplier A	1,00	2,00	2,00	3,00	8,00	2,00
Supplier B	0,33	1,00	3,00	1,00	5,33	1,33
Supplier C	0,50	0,33	1,00	2,00	3,83	0,96
Supplier D	0,33	1,00	0,50	1,00	2,83	0,71

Calculating Ratio - C4							
Criteria C4	Supplier A	Supplier B	Supplier C	Supplier D	PW	Matrix Results	Divided Results
Supplier A	1,00	2,00	3,00	2,00	2,00	8,96	4,48
Supplier B	0,50	1,00	1,00	3,00	1,33	5,42	4,06
Supplier C	0,33	1,00	1,00	2,00	0,96	4,38	4,57
Supplier D	0,50	0,20	0,50	1,00	0,71	2,45	3,46
Count Maks - C4							
Total	4,14						
Count Consistency Index - C4							
Total	0,05						
Consistency Ratio - C4							
Total	0,05	Consistent					

F. Costs Calculation

- Intensity of Interest

Intensity of Interest				
Criteria	C1	C2	C3	C4
C1	1/1	1/4	1/5	1/6
C2	4/1	1/1	1/3	1/4
C3	5/1	3/1	1/1	1/2
C4	6/1	4/1	2/1	1/1

- Normalization

Normalization						
Criteria	C1	C2	C3	C4		
C1	1,00	0,25	0,20	0,17		
C2	4,00	1,00	0,33	0,25		
C3	5,00	3,00	1,00	0,50		
C4	6,00	3,00	2,00	1,00		
Total	16,00	7,25	3,53	1,92		
Criteria	C1	C2	C3	C4	Total	PW
C1	0,06	0,03	0,06	0,09	0,24	0,06
C2	0,25	0,14	0,09	0,13	0,61	0,15
C3	0,31	0,41	0,28	0,26	1,27	0,32
C4	0,38	0,41	0,57	0,52	1,88	0,47

- Calculating Ratio

Calculating Ratio							
Criteria	C1	C2	C3	C4	PW	Matrix Results	Divided Results
C1	1,00	3,00	3,00	5,00	0,06	3,82	63,49
C2	0,33	1,00	1,00	3,00	0,15	1,90	12,39
C3	0,33	1,00	1,00	3,00	0,32	1,90	5,98
C4	0,20	0,33	0,33	1,00	0,47	0,64	1,36

- Count Maks

Count Maks		
Total	20,80	

- Count Consistency Index

Count Consistency Index		
Total	0,90	

- Consistency Ratio

Consistency Ratio		
Total	0,05	Consistent

- Consistency Ratio for C1

Intensity of Interest - C1						
Criteria C1	Supplier A	Supplier B	Supplier C	Supplier D		
Supplier A	1/1	2/1	4/1	3/1		
Supplier B	1/2	1/1	3/1	2/1		
Supplier C	1/4	1/3	1/1	2/1		
Supplier D	1/3	1/2	1/2	1/1		
Normalization - C1						
Criteria C1	Supplier A	Supplier B	Supplier C	Supplier D		
Supplier A	1,00	2,00	4,00	3,00		
Supplier B	0,50	1,00	3,00	2,00		
Supplier C	0,25	0,33	1,00	2,00		
Supplier D	0,33	0,33	0,50	1,00		
Total	2,08	3,67	8,50	8,00		
Criteria C1	Supplier A	Supplier B	Supplier C	Supplier D	Total	PW
Supplier A	0,48	0,55	0,47	0,38	1,87	0,47
Supplier B	0,24	0,27	0,35	0,25	1,12	0,28
Supplier C	0,12	0,09	0,12	0,25	0,58	0,14
Supplier D	0,16	0,09	0,06	0,13	0,43	0,11

Calculating Ratio - C1							
Criteria C1	Supplier A	Supplier B	Supplier C	Supplier D	PW	Matrix Results	Divided Results
Supplier A	1,00	1,00	3,00	2,00	0,47	1,40	2,99
Supplier B	1,00	1,00	2,00	1,00	0,28	1,14	4,10
Supplier C	0,50	0,50	1,00	1,00	0,14	0,63	4,33
Supplier D	0,50	1,00	1,00	1,00	0,11	0,77	7,05
Count Maks - C1							
Total	4,20						
Count Consistency Index - C1							
Total	0,07						
Consistency Ratio - C1							
Total	0,07	Consistent					

- Consistency Ratio for C2

Intensity of Interest - C2						
Criteria C2	Supplier A	Supplier B	Supplier C	Supplier D		
Supplier A	1/1	3/1	2/1	4/1		
Supplier B	1/3	1/1	2/1	3/1		
Supplier C	1/2	1/2	1/1	2/1		
Supplier D	1/4	1/3	1/2	1/1		
Normalization - C2						
Criteria C2	Supplier A	Supplier B	Supplier C	Supplier D		
Supplier A	1,00	3,00	2,00	4,00		
Supplier B	0,33	1,00	2,00	3,00		
Supplier C	0,50	0,50	1,00	2,00		
Supplier D	0,25	0,33	0,50	1,00		
Total	2,08	4,83	5,50	10,00		
Criteria C2	Supplier A	Supplier B	Supplier C	Supplier D	Total	PW
Supplier A	0,48	0,62	0,36	0,40	1,86	0,47
Supplier B	0,16	0,21	0,36	0,30	1,03	0,26
Supplier C	0,24	0,10	0,18	0,20	0,73	0,18
Supplier D	0,12	0,07	0,09	0,10	0,38	0,09

Calculating Ratio - C2							
Criteria C2	Supplier A	Supplier B	Supplier C	Supplier D	PW	Matrix Results	Divided Results
Supplier A	1,00	2,00	3,00	2,00	0,47	1,72	3,68
Supplier B	0,50	1,00	1,00	3,00	0,26	0,96	3,71
Supplier C	0,33	1,00	1,00	2,00	0,18	0,78	4,33
Supplier D	0,50	0,20	0,50	1,00	0,09	0,47	4,95
Count Maks - C2							
Total	4,20						
Count Consistency Index - C2							
Total	0,07						
Consistency Ratio - C2							
Total	0,07	Consistent					

- Consistency Ratio for C3

Intensity of Interest - C3						
Criteria C3	Supplier A	Supplier B	Supplier C	Supplier D		
Supplier A	1/1	3/1	2/1	5/1		
Supplier B	1/3	1/1	4/1	3/1		
Supplier C	1/2	1/4	1/1	2/1		
Supplier D	1/5	1/3	1/2	1/1		
Normalization - C3						
Criteria C3	Supplier A	Supplier B	Supplier C	Supplier D		
Supplier A	1,00	3,00	2,00	5,00		
Supplier B	0,33	1,00	4,00	3,00		
Supplier C	0,50	0,25	1,00	2,00		
Supplier D	0,20	0,33	0,50	0,20		
Total	2,03	4,58	7,50	10,20		
Criteria C3	Supplier A	Supplier B	Supplier C	Supplier D	Total	PW
Supplier A	0,49	0,66	0,27	0,45	1,87	0,47
Supplier B	0,16	0,22	0,53	0,27	1,18	0,30
Supplier C	0,25	0,05	0,13	0,18	0,61	0,15
Supplier D	0,10	0,07	0,07	0,09	0,33	0,08

Calculating Ratio - C3							
Criteria C3	Supplier A	Supplier B	Supplier C	Supplier D	PW	Matrix Results	Divided Results
Supplier A	1,00	2,00	3,00	2,00	0,47	1,68	3,59
Supplier B	0,50	1,00	1,00	3,00	0,30	0,93	3,15
Supplier C	0,33	1,00	1,00	2,00	0,15	0,77	5,04
Supplier D	0,50	0,20	0,50	1,00	0,08	0,45	5,47
Count Maks - C3							
Total	4,24						
Count Consistency Index - C3							
Total	0,08						
Consistency Ratio - C3							
Total	0,09	Consistent					

- Consistency Ratio for C4

Intensity of Interest - C4						
Criteria C4	Supplier A	Supplier B	Supplier C	Supplier D		
Supplier A	1/1	2/1	4/1	3/1		
Supplier B	1/2	1/1	3/1	2/1		
Supplier C	1/4	1/3	1/1	2/1		
Supplier D	1/3	1/2	1/2	1/1		
Normalization - C4						
Criteria C4	Supplier A	Supplier B	Supplier C	Supplier D		
Supplier A	1,00	2,00	4,00	3,00		
Supplier B	0,50	1,00	3,00	2,00		
Supplier C	0,25	0,33	1,00	2,00		
Supplier D	0,33	0,50	0,50	1,00		
Total	2,08	3,83	8,50	8,00		
Criteria C4	Supplier A	Supplier B	Supplier C	Supplier D	Total	PW
Supplier A	0,48	0,52	0,47	0,38	1,85	0,46
Supplier B	0,24	0,26	0,35	0,25	1,10	0,28
Supplier C	0,12	0,09	0,12	0,25	0,58	0,15
Supplier D	0,16	0,13	0,06	0,13	0,48	0,12

Calculating Ratio - C4							
Criteria C4	Supplier A	Supplier B	Supplier C	Supplier D	PW	Matrix Results	Divided Results
Supplier A	1,00	2,00	3,00	2,00	0,46	1,69	3,65
Supplier B	0,50	1,00	1,00	3,00	0,28	1,01	3,68
Supplier C	0,33	1,00	1,00	2,00	0,15	0,81	5,61
Supplier D	0,50	0,20	0,50	1,00	0,12	0,48	3,99
Count Maks - C4							
Total	4,20						
Count Consistency Index - C4							
Total	0,07						
Consistency Ratio - C4							
Total	0,07	Consistent					

G. Final Results

DECISION MAKING BENEFITS					
	C1 (Benefit)	C2 (Benefit)	C3 (Benefit)	C4 (Benefit)	Evaluation Weight
Priority Weight	0,52	0,20	0,20	0,08	
Supplier A	0,36	0,43	2,25	2,00	0,88
Supplier B	0,28	0,25	1,33	1,33	5,86
Supplier C	0,16	0,20	0,96	0,96	2,65
Supplier D	0,20	0,12	0,71	0,71	1,42
DECISION MAKING COSTS					
	C1 (Cost)	C2 (Cost)	C3 (Cost)	C4 (Cost)	Evaluation Weight
Priority Weight	0,06	0,15	0,32	0,47	
Supplier A	0,47	0,47	0,47	0,46	0,47
Supplier B	0,28	0,26	0,30	0,28	0,52
Supplier C	0,14	0,18	0,15	0,15	0,17
Supplier D	0,11	0,09	0,08	0,12	0,06

DECISION MAKING FINAL RESULTS			
Alternative	Benefit Value	Cost Value	Final Value
Supplier A	0,88	0,47	1,90
Supplier B	5,86	0,52	11,19
Supplier C	2,65	0,17	15,31
Supplier D	1,42	0,06	23,04
RANK RESULTS			
No	Alternative	Score	
1	Supplier D	23,04	
2	Supplier C	15,31	
3	Supplier B	11,19	
4	Supplier A	1,90	

H. Conclusions

Based on the analysis using the Analytical Hierarchy Process (AHP) method, the calculation results show that Supplier D has the highest score, which is 23.04, so it is the best choice as the main supplier. The ranking order of suppliers based on the final score is as follows:

1. **Supplier D – 23,04** (Rank 1)
2. **Supplier C – 15,31** (Rank 2)
3. **Supplier B – 11,19** (Rank 3)
4. **Supplier A – 1,90** (Rank 4)

From these results, it can be concluded that Supplier D is superior in meeting benefit criteria such as raw material quality, delivery speed, reputation, and after-sales service compared to other suppliers. In addition, Supplier D is also more optimal in cost aspects, such as price per unit, shipping costs, minimum order quantity, and risk of delays.

In contrast, Supplier A has the lowest score (1.90), indicating that it falls short of meeting the criteria, both in terms of service quality and cost efficiency.

By considering all the factors that have been calculated using the AHP method, Supplier D is recommended as the main choice to ensure the supply of quality, efficient raw materials, and in accordance with the company's needs.

Here the link for excel results:

https://docs.google.com/spreadsheets/d/1PkLvrmt3uA1oxMLE0_ANVGh2ppPmHAAtfkQKhuStDGs/edit?usp=sharing